MATH2921 Final Exam 2022

Extended Answer Questions

Extended Answer Question 1 (5 marks)

For the unknown $u(x,t)$ over $\mathbb{R} \times \mathbb{R}$, solve

$u_{tt} - 9u_{xx} = 0, u(x,0) = \cos(2x), u_t(x,0) = \sin x.$

Extended Answer Question 2 (11 marks)

For the unknown $u(x,y)$ over $[0,\pi] \times [0,2\pi]$, solve

$u_{xx} + u_{yy} = 0, \;\; u(x,0) = u_x(0,y) = u_x(\pi,y) = 0, \;\; u(x,2\pi) = \cos x.$

Extended Answer Question 3 (6 marks)

For the unknown $y(x)$, solve

$y'' - y' - 2 = 0.$

Extended Answer Question 4 (10 marks)

Consider $S = \{(x,y,z) \mid x^2 + y^2 + z^2 = 25, \; 0 \le z \le 3\}$ and the vector field $\overrightarrow{F}(x,y,z) = (y,z,x)$.

Applying Stokes' Theorem or not, calculate

$\iint_S (\nabla \times \overrightarrow{F}) \cdot \overrightarrow{n} \, dS$

where $\overrightarrow{n}$ points in the direction to leave the origin.

Extended Answer Question 5 (11 marks)

Let $H(x,y) = x^2 - xy + y^2$. Consider the vector field $\overrightarrow{F}(x,y) = (H_y, H_x)$. The curve $\overrightarrow{r}(t) = \big(x(t), y(t)\big)$ satisfies

$\overrightarrow{r}'(t) = \overrightarrow{F}\big(\overrightarrow{r}(t)\big).$

(i) Write down the 1st order ODE system for the unknown $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ from $\overrightarrow{r}(t)$.

(ii) Solve this ODE system, and get the curve $\overrightarrow{r}(t)$ starting from $(1,1)$, in other words, $\overrightarrow{r}(0) = (1,1)$.

Extended Answer Question 6 (7 marks)

When optimising a function $f(x, y)$ under the constraint $F(x, y) = 0$, one can consider the local property of each critical point as follows. By Lagrange Multiplier Method, a critical point (a, b) satisfies

$$\nabla(f + \lambda F)(a, b) = (0, 0), \quad F(a, b) = 0$$

for some proper constant λ. Near (a, b), we assume $F(x, y) = 0$ implicitly gives $y = g(x)$. So we can optimise the one-variable function $f\big(x, g(x)\big)$. Prove the following statements:

(i) $x = a$ is a critical point for $f\big(x, g(x)\big)$.

(ii) The second derivative of $f\big(x, g(x)\big)$ at $x = a$ is equal to

$$(1, g') \cdot D^2(f + \lambda F) \cdot \begin{bmatrix} 1 \\ g' \end{bmatrix}$$

for $(x, y) = (a, b)$, where $D^2(f + \lambda F)$ is the Hessian matrix for $f + \lambda F$ and "$\cdot$" is matrix multiplication.